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③ Using fibonacci sequence use $F(5)$ to illustrate dynamic programming

i) Memoization (Top down)

Memoization uses recursion and stores previous computed results

for the fibo of 5

$$F(5) = F(4) + F(3)$$

Computing $F(4) = F(3) + F(2)$

$$F(3) = F(2) + F(1)$$

$$F(2) = F(1) + F(0)$$

but we have $F(0) = 0$ and $F(1) = 1$

Hence substituting in the value,

$$F(2) = 1 \quad F(3) = 2 \quad F(4) = 3 \quad F(5) = 5$$

$$\underline{\underline{F(5) = 5}}$$

1.) Tabulation (Bottom up)

Tabulation Method of dynamic programming works by starting from the smallest values and by filling it solve the recursion problem.

Starting from the base upwards

$$F(2) = F(1) + F(0) = 1$$

$$F(3) = F(2) + F(1) = 2$$

$$F(4) = F(3) + F(2) = 3$$

$$F(5) = F(4) + F(3) = 5$$

$$\underline{\underline{F(5) = 5}}$$